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Read the following passage.

ATP is essential for muscle contraction. The concentration of ATP in skeletal muscle is approximately $5 \times 10^{-3} \text{ mmol g}^{-1}$. During maximum exercise, ATP in skeletal muscle is used at a rate of approximately $3.7 \text{ mmol kg}^{-1} \text{ s}^{-1}$. If ATP was not resynthesised, maximum exercise would last a short time.

Some studies have suggested that taking creatine supplements can improve muscle performance during intense short-term exercise. Eating more carbohydrate, known as 'carbohydrate loading', for a few days before exercise can improve muscle performance when exercising for a long time. 5

During aerobic exercise, calcium ions in muscle cells have several roles. They activate the inclusion of glucose transport proteins, GLUT4, in the cell-surface membrane and the inclusion of fatty acid transport proteins, CD36, in mitochondrial membranes. An increase in muscle activity increases the concentration of carbon dioxide in the blood. This causes an increase in heart rate. 10

Use the information in the passage and your own knowledge to answer the following questions.

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ATP is essential for muscle contraction (line 1).

Describe the roles of ATP in muscle contraction.

[2 marks]

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Calculate how long maximum exercise would last if ATP was not resynthesised (lines 3–4).

[1 mark]

Answer _____ s

[5 marks]

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Suggest and explain how GLUT4 and CD36 transport proteins (lines 10–11) are beneficial during exercise.

[3 marks]

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An increase in muscle activity causes an increase in heart rate (lines 12–14).

Describe and explain how.

[4 marks]

END OF QUESTIONS

[illegible]

[illegible]